

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

**COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems**
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: Total Marks: Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Edge Computing over Satellite Networks for Future Wireless Systems

Centralized cloud networks cannot handle future traffic (500+ billion connected devices by 2030), causing heavy network congestion and high delays. Edge Computing over Satellite Networks solves this by putting processing power directly on Low Earth Orbit (LEO) satellites in 6G systems. This creates a 3D space-air-ground network architecture that delivers sub-5-millisecond low latency and global coverage for remote, maritime, and aerial areas.

Using spaceborne Multi-Access Edge Computing (MEC) lets satellites process tasks in the sky instead of sending data back to ground servers. To manage fast-moving satellites and changing links, complex resource sharing can be simplified using an optimization model (OEDMA) under the CROWN framework to easily assign tasks across moving satellite nodes.

Key system parameters for satellite edge computing include offloading decisions (local vs. satellite processing), strict task delay limits, satellite CPU power, dynamic bandwidth, and energy limits. Satellite selection relies on metrics like Primacy (node rank), Supremacy (task suitability), and Sway (occupancy time). Integrated Sensing and Communications (ISAC) tracks orbital positions in real time, while Proof of Reputation keeps data sharing secure. Satellite edge computing is vital for 6G, though key challenges remain in satellite battery limits, fast handovers, and space cybersecurity.



Edge computing in future wireless networks: A comprehensive evaluation and vision for 6G and beyond

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Abstract

Future Internet aims to function as a neutral in-network storage and computation platform, essential for enabling 6G and beyond wireless use cases. Information-Centric Networking and Edge Computing are key paradigms driving this vision by offering diversified services with fast response times across heterogeneous networks. This approach requires effective coordination to dynamically utilize resources like links, storage, and computation in near real-time within a non-homogenous and distributed computing environment. Additionally, networks must be aware of resource availability and reputational information to manage unknown and partially observed dynamic systems, ensuring the desired Quality of Experience (QoE). This paper provides a comprehensive evaluation of edge computing technologies, starting with an introduction to its architectural frameworks. We examine contemporary research on essential aspects such as resource allocation, computation delegation, data administration, and network management, highlighting existing research gaps. Furthermore, we explore the synergy between edge computing and 5G, and discuss advancements in 6G that enhance solutions through edge computing. Our study emphasizes the importance of integrating edge computing in future considerations, particularly regarding sustainable energy and standards.

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Keywords: Wireless communication; 5G/6G; IoT/6G; Communication; Latency; Edge computing; CROWN

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